

CHANGSHA, China

WMO: 576870

Lat: 28.1095N

Lon: 112.7880E

Elev: 120

StdP: 99.89

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.2	(c) -0.1	(d) -8.6	(e) 1.8	(f) 6.2	(g) -5.9	(h) 2.3	(i) 5.9	(j) 7.6	(k) 4.3	(l) 6.9	(m) 5.0	(n) 2.6	(o) 340	(p) 0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.7	(c) 36.4	(d) 26.0	(e) 35.3	(f) 26.0	(g) 34.2	(h) 25.9	(i) 27.8	(j) 32.6	(k) 27.3	(l) 32.1	(m) 26.8	(n) 31.6	(o) 2.8	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 26.7	(b) 22.6	(c) 30.1	(d) 26.1	(e) 21.8	(f) 29.8	(g) 25.6	(h) 21.1	(i) 29.4	(j) 90.1	(k) 32.7	(l) 87.6	(m) 32.0	(n) 85.4	(o) 31.8	(p) 30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.9	(b) 5.9	(c) 5.2	(e) -2.6	(f) 38.0	(g) 1.2	(h) 1.0	(i) -3.5	(j) 38.8	(k) -4.2	(l) 39.4	(m) -4.9	(n) 40.0	(o) -5.8	(p) 40.7		
(a) 6.9	(b) 5.9	(c) 5.2	(e) -3.5	(f) 28.5	(g) 1.1	(h) 0.8	(i) -4.3	(j) 29.1	(k) -5.0	(l) 29.6	(m) -5.6	(n) 30.0	(o) -6.4	(p) 30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	5.5	8.0	12.5	18.4	23.0	26.3	29.7	28.9	24.7	19.4	13.6	7.9
	DBStd	8.97	3.58	4.53	4.63	4.36	3.32	2.91	2.76	2.94	3.55	3.54	4.03	3.19
	HDD10.0	349	146	85	25	1	0	0	0	0	0	0	13	79
	HDD18.3	1435	398	290	190	52	3	0	0	0	2	29	148	323
	CDD10.0	3351	7	29	102	254	402	489	612	584	442	293	122	14
	CDD18.3	1394	0	1	8	55	147	239	354	326	194	63	7	0
	CDH23.3	14483	1	9	56	367	1088	2311	4644	3966	1682	328	33	0
CDH26.7	6534	0	1	12	97	362	922	2468	1931	663	75	4	0	
Wind	WSAvg	2.3	2.4	2.5	2.4	2.4	2.2	2.1	2.4	2.4	2.4	2.3	2.3	2.4
Precipitation	PrecAvg	1488	77	88	146	183	210	229	153	115	83	65	86	54
	PrecMax	1778	175	192	360	275	446	495	379	311	219	156	208	136
	PrecMin	979	16	13	84	81	52	123	11	25	3	3	12	0
	PrecStd	223	39	39	60	57	86	93	98	62	55	36	58	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	24.7	28.2	31.7	33.7	35.1	37.7	38.0	35.9	31.4	26.8	19.3
		MCWB	12.4	16.9	20.7	21.9	23.8	25.8	26.0	25.5	25.3	21.5	18.4	12.3
	2%	DB	15.3	20.2	24.5	29.4	32.1	34.0	36.6	36.4	33.9	28.7	23.8	16.7
		MCWB	9.8	14.0	17.6	21.1	23.1	25.8	26.1	25.9	25.4	20.5	17.0	10.6
	5%	DB	12.9	17.2	22.0	27.3	30.4	32.8	35.6	35.1	32.2	26.7	21.6	14.6
		MCWB	8.5	12.2	15.9	20.0	22.4	25.5	26.0	26.2	24.9	19.3	15.4	9.5
	10%	DB	10.7	14.4	19.5	25.3	28.7	31.3	34.6	33.7	30.5	25.0	19.7	12.7
		MCWB	7.5	10.4	14.7	19.4	22.1	25.1	26.0	26.2	24.0	18.8	14.5	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	17.3	21.3	23.9	25.9	27.4	28.3	28.7	27.4	23.9	20.1	14.3
		MCDB	17.8	23.7	27.6	30.1	30.9	32.7	33.4	33.4	32.4	28.9	24.6	16.9
	2%	WB	11.0	15.0	18.8	22.6	24.9	26.7	27.7	28.0	26.5	22.2	18.2	11.9
		MCDB	13.6	19.3	22.9	27.3	29.1	31.7	32.8	32.6	31.3	25.8	22.0	14.6
	5%	WB	9.0	12.8	17.1	21.5	24.0	26.2	27.2	27.4	25.8	21.1	16.8	10.5
		MCDB	11.9	16.0	20.6	25.4	28.0	31.0	32.3	32.0	30.3	24.9	20.0	13.2
	10%	WB	7.7	10.9	15.4	20.3	23.2	25.7	26.8	26.9	25.1	20.1	15.5	9.3
		MCDB	10.1	13.9	18.5	24.2	27.0	30.1	31.8	31.4	29.2	23.0	18.3	12.0
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.5	6.2	6.8	6.9	6.2	6.7	6.5	6.7	6.6	6.4	5.9
		MCDBR	10.1	10.7	10.8	10.5	9.9	8.2	8.1	8.4	9.1	9.8	10.3	10.5
		MCWBR	5.5	5.6	5.2	4.4	3.5	2.7	2.0	2.1	2.7	3.3	4.4	5.2
	5% WB	MCDBR	8.4	9.5	9.3	8.9	7.8	7.1	7.6	7.7	7.9	7.9	7.9	7.9
		MCWBR	5.1	5.5	5.4	4.7	3.7	2.9	2.5	2.7	2.8	3.6	4.3	4.4
Clear-Sky Solar Irradiance	taub	0.705	0.745	0.790	0.772	0.700	0.683	0.604	0.679	0.710	0.751	0.700	0.681	
	taud	1.475	1.441	1.399	1.473	1.633	1.684	1.877	1.681	1.597	1.473	1.534	1.536	
	Ebn at Noon	533	561	575	610	659	668	721	663	621	549	533	525	
	Edh at Noon	260	290	319	304	259	245	202	244	258	276	243	235	
All-Sky Solar Radiation	RadAvg	1.68	2.01	2.69	3.36	3.84	4.12	5.28	4.55	3.77	2.95	2.34	1.93	
	RadStd	0.36	0.56	0.30	0.43	0.57	0.49	0.54	0.39	0.41	0.46	0.40	0.43	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.39	N/A	-2.20	N/A	-0.50	-0.51	N/A	N/A	+165	N/A
(47) Regional (6 neighbors)		+0.24	N/A	N/A	+0.44	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page